Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

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1. CUSTOMER SEGMENT(S) [CS]

- Government organizations
- Policy makers
- Researchers and analysts
- Educational institutions
- NGOs
- Students studying human development
- General users interested in HDI analysis

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6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES **[CL]**

- Limited technical knowledge.
- Limited access to reliable datasets.
- Budget constraints.
- Limited computing resources.
- Lack of awareness of machine learning tools.

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5. AVAILABLE SOLUTIONS PROS & CONS **[AS]**

Pros

- UNDP reports provide authentic information.
- Statistical tools offer detailed analysis.
- Existing ML methods are accurate.

Cons

- Manual analysis is slow.
- Existing systems require technical knowledge.
- Many tools are not user-friendly.
- Limited instant prediction capability.

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2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Difficulty in estimating HDI manually.

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9. PROBLEM ROOT / CAUSE [RC]

- Manual HDI calculation is complex.

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7. BEHAVIOR + ITS INTENSITY **[BE]**

- Frequently searches for HDI information.

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- Time-consuming analysis of multiple development indicators.
- Lack of quick decision support.
- Limited access to simple prediction tools.
- Frequent need for updated development assessment.

- Multiple development indicators must be analyzed.
- Lack of accessible prediction systems.
- Limited availability of automated tools.
- Increasing need for quick development assessment.

- Compares country development levels.
- Uses online prediction tools.
- Requires quick and accurate results.
- High demand during research and policy planning.

3. TRIGGERS TO ACT [TR]

- Need to estimate a country's HDI quickly.
- Government policy planning.
- Academic research and projects.
- Development comparison between countries.
- Data-driven decision making.

4. EMOTIONS

BEFORE / AFTER **[EM]**

Before

- Confused
- Time-consuming calculations
- Uncertain about results

After

- Confident
- Faster analysis
- Reliable prediction
- Better decision making

10. YOUR SOLUTION [SL]

Develop a **Machine Learning-based Human Development Index Prediction System** using **Flask** that predicts the HDI score based on **Country, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income per Capita**. The system instantly classifies the country into **Low, Medium, High, or Very High HDI**, providing a simple, accurate, and user-friendly web application for researchers, students, and policymakers.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Web application
- GitHub
- Research websites
- Educational portals

OFFLINE

- Universities
- Government offices
- Research organizations
- Libraries